6.1 Deformation: Stress & Strain

Question Paper

Course	CIE A Level Physics
Section	6. Deformation of Solids
Topic	6.1 Deformation: Stress & Strain
Difficulty	Easy

Time allowed: 50

Score: /35

Percentage: /100

Question 1a

Forces can change the motion of a body, or cause deformation.

State

(i)
the definition of *deformation*.

[1]

(ii)
the name of the force which stretches an object.

[1]

(iii)
the name of the force which squashes an object.

[1]

[3 marks]

Question 1b

Hooke's Law can be applied to objects which are stretched or compressed.

State the equation for Hooke's Law, defining any terms used.

[2 marks]

Question 1c

A spring is suspended from a stand as shown in Fig. 1.1 and has a load of 5.0 N applied to it. The spring extends by 4.2 cm.

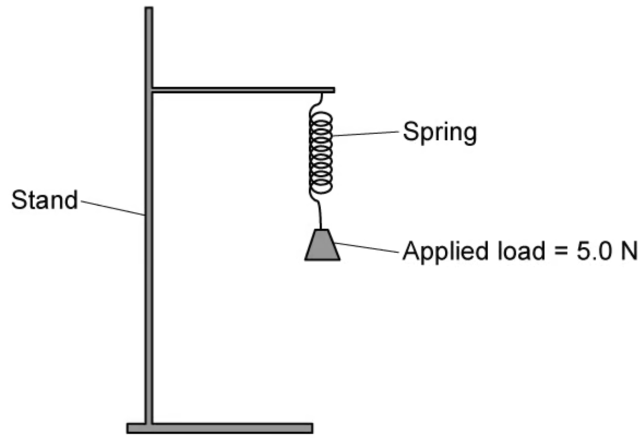


Fig. 1.1

Calculate the spring constant, expressing your answer in S.I. units.

[3 marks]

Question 1d

Another, thicker spring is attached to a base so that it stands upright, as shown in Fig. 1.2.

A force of 6.3 N pushes down on the spring which compresses it by 2.4 cm.

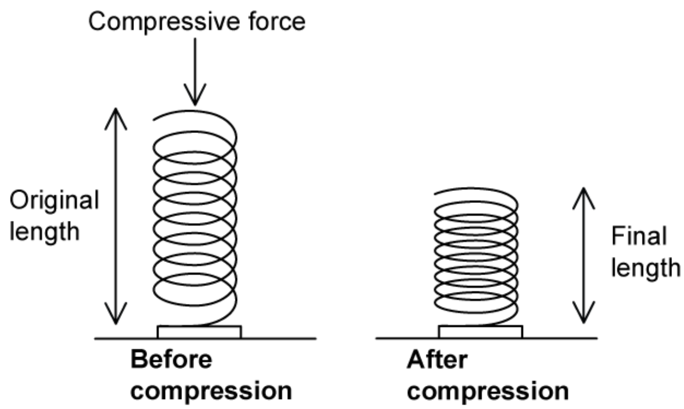


Fig. 1.2.

Calculate the spring constant, expressing your answer in S.I. units.

[1 mark]

Question 2a

Define

(i)
tensile stress

[1]

(ii)
tensile strain

[1]

(iii)
the Young modulus.

[1]

[3 marks]

Question 2b

Fig. 1.1 shows stress-strain graphs for two different metal wires X and Y.

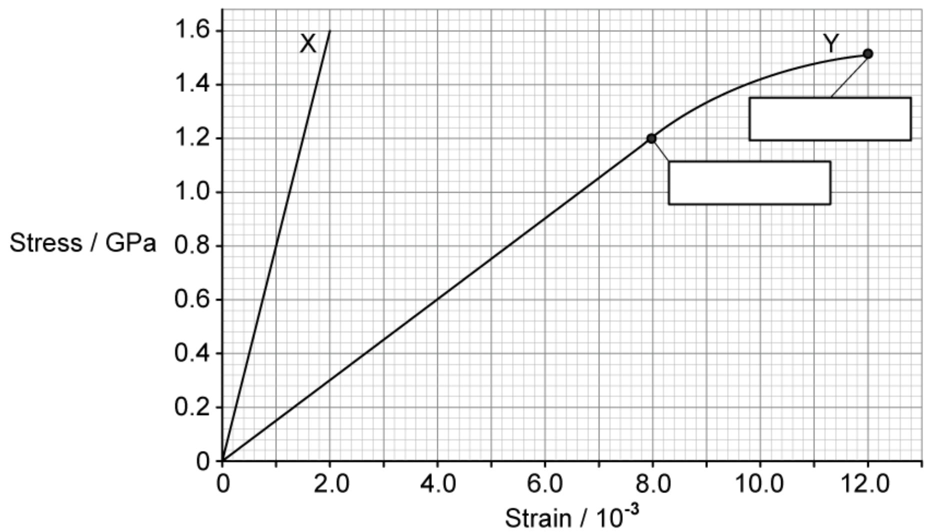


Fig. 1.1

Fill in the missing labels on Fig. 1.1 and state the definitions of these terms.

[4 marks]

Question 2c

State and explain which wire, X or Y, has the greater Young modulus.

[2 marks]

Question 2d

Use the graph in Fig. 1.1 to determine

(i)

the Young modulus of wire X and state the unit.

[3]

(ii)

the extension of wire Y when the stress is 1.2 GPa, and state the unit.

original length of wire Y = 2.0 m

[3]

[6 marks]

Question 3a

A student is carrying out an experiment to determine the Young modulus of a wire. The student sets up the apparatus as shown in Fig. 1.1.

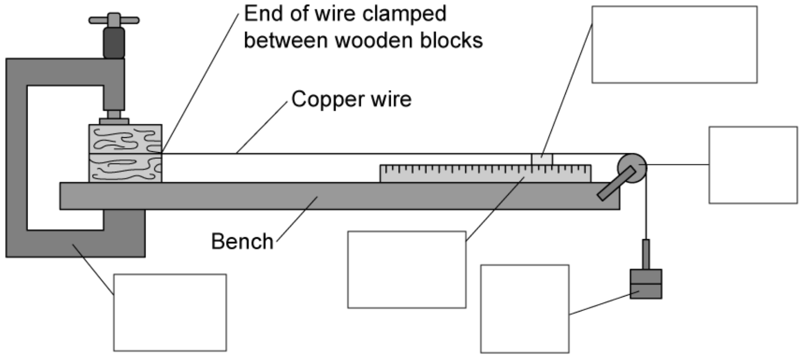


Fig. 1.1

Fill in the missing labels for the apparatus in Fig. 1.1.

[5 marks]

Question 3b

State and explain a safety consideration for this experiment.

[2 marks]

Question 3c

On the axis of Fig. 1.2 below, draw a graph to show the expected variation of strain with stress for this experiment.

Assume that the wire does not stretch beyond the limit of proportionality.

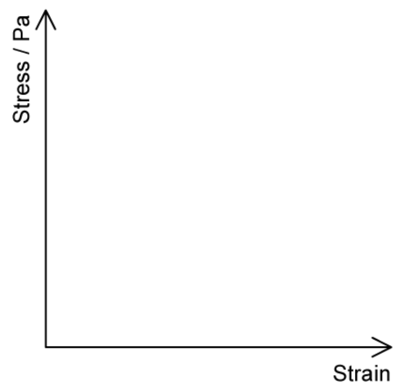


Fig. 1.2

[2 marks]

Question 3d

Explain how the student could use the graph in Fig. 1.2 to determine the Young modulus of the wire.

[2 marks]

Model Answers: Easy

1a

(i) Deformation is defined as:

- Changing the size / shape (of an object); [1 mark]

(ii) The force which stretches an object is called:

- Tensile (stress / force); [1 mark]

(iii) The force which squashes an object is called:

- Compressive (stress / force); [1 mark]

[Total: 3 marks]

1b

Hooke's Law can be written as:

- $F = kx$ [1 mark]
- Where F = force, k = spring constant and x is the extension; [1 mark]

[Total: 2 marks]

1c

Calculate the spring constant, expressing your answer in S.I. units:

List the known quantities:

- Load, F = 5.0 N
- Extension, x = 4.2 cm = 0.042 m [1 mark]

Use the equation for the spring constant and calculate:

- Hooke's law: $F = kx \quad \Rightarrow \quad k = \frac{F}{x}$
- Spring constant: $k = \frac{5.0}{0.042}$ [1 mark]

- Spring constant: $k = 120 \text{ N m}^{-1} \text{ (2 s.f.)}$ [1 mark]

[Total: 3 marks]

1d

Calculate the spring constant, expressing your answer in S.I. units:

List the known quantities:

- Compressive force, F = 6.3 N
- Compression, x = 2.4 cm = 0.024 m [1 mark]

Use the equation for the spring constant and calculate:

- Spring constant: $k = \frac{F}{x} = \frac{6.3}{0.024}$ [1 mark]

- Spring constant: $k = 260 \text{ Nm}^{-1} \text{ (2 s.f.)}$ [1 mark]

[Total: 3 marks]

Hooke's Law questions are nearly always asked about springs under tension, being stretched. However, be aware that exactly the same equation also applies to compressive forces and that it's used in the same way.

2a

(i) Tensile stress is defined as:

- The ratio of the force to the cross-sectional area
  OR
- Stress = $\frac{F}{A}$, where F = force and A = cross-sectional area; [1 mark]

(ii) Tensile strain is defined as:

- The ratio of extension to initial length;
  OR
- Strain = $\frac{x}{L}$, where x = extension and L = original length; [1 mark]

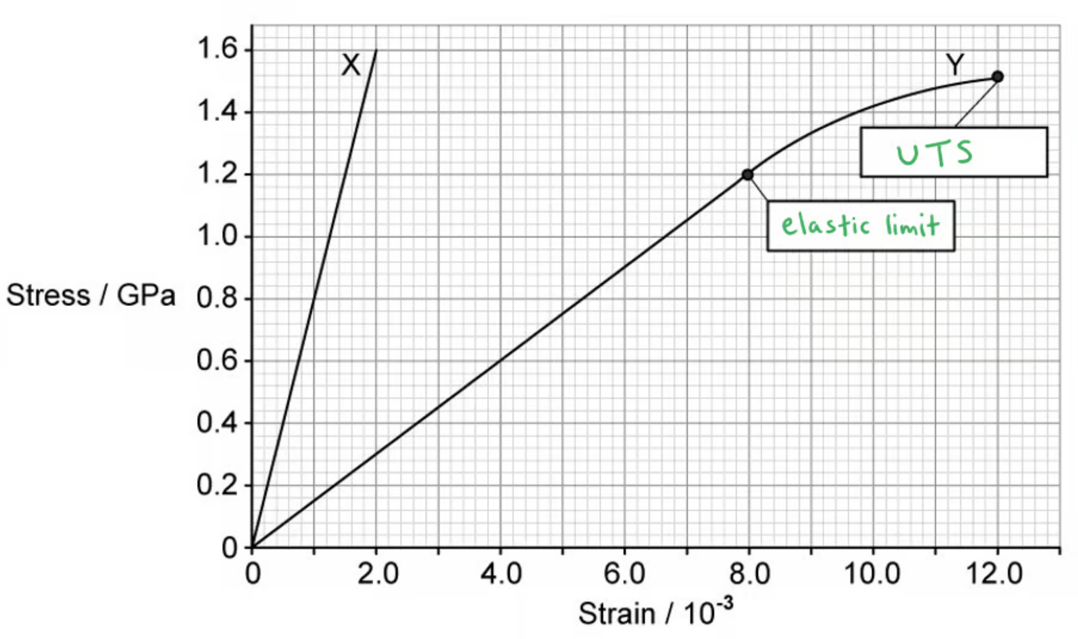
(iii) The Young modulus is defined as:

- The ratio of stress to strain;
  OR
- Young modulus = $\frac{stress}{strain} \left(= \frac{FL}{Ax} \right)$ [1 mark]

[Total: 3 marks]

2b

The correct labels are:



- Elastic limit (as shown); [1 mark]
- Ultimate tensile strength / UTS / breaking stress (as shown); [1 mark]

Elastic limit is defined as:

- The point beyond which the wire will not return to its original length / shape / size when the load / force is removed; [1 mark]

Ultimate tensile strength / breaking stress is defined as:

- The maximum stress the wire is able to support / before it breaks; [1 mark]

[Total: 4 marks]

2c

The wire with the greater Young modulus is

- Wire X; [1 mark]

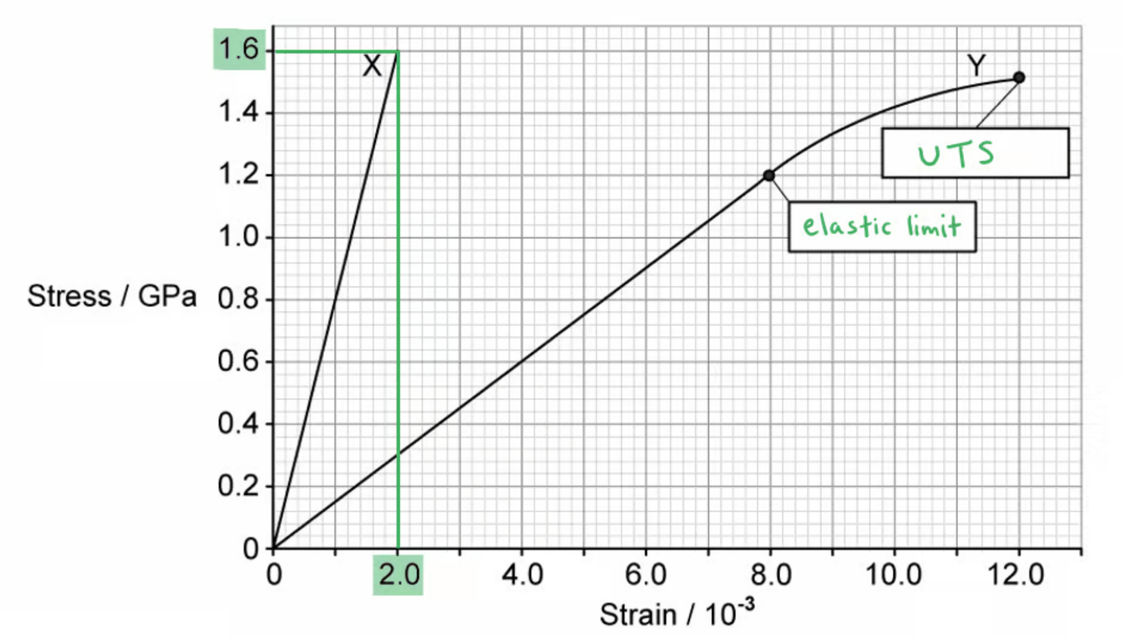
Because:

- X has the steeper gradient / larger value of stress/strain (before breaking); [1 mark]

[Total: 2 marks]

2d

(i) Determine the Young modulus of wire X:



- (From graph) when stress = 1.6 GPa, strain = 2.0 × 10^{-3} [1 mark]
- Young modulus = gradient of the stress-strain graph
- Young modulus = $\frac{stress}{strain}$ = $\frac{1.6 \times 10^9}{2.0 \times 10^{-3}}$ [1 mark]
- Young modulus = 8.0 × 10^{11} Pa **OR** 800 GPa [1 mark]

(ii) Determine the extension of wire Y when the stress is 1.2 GPa:

- The maximum stress the wire is able to support / before it breaks; [1 mark]

[Total: 4 marks]

2c

The wire with the greater Young modulus is

- Wire X; [1 mark]

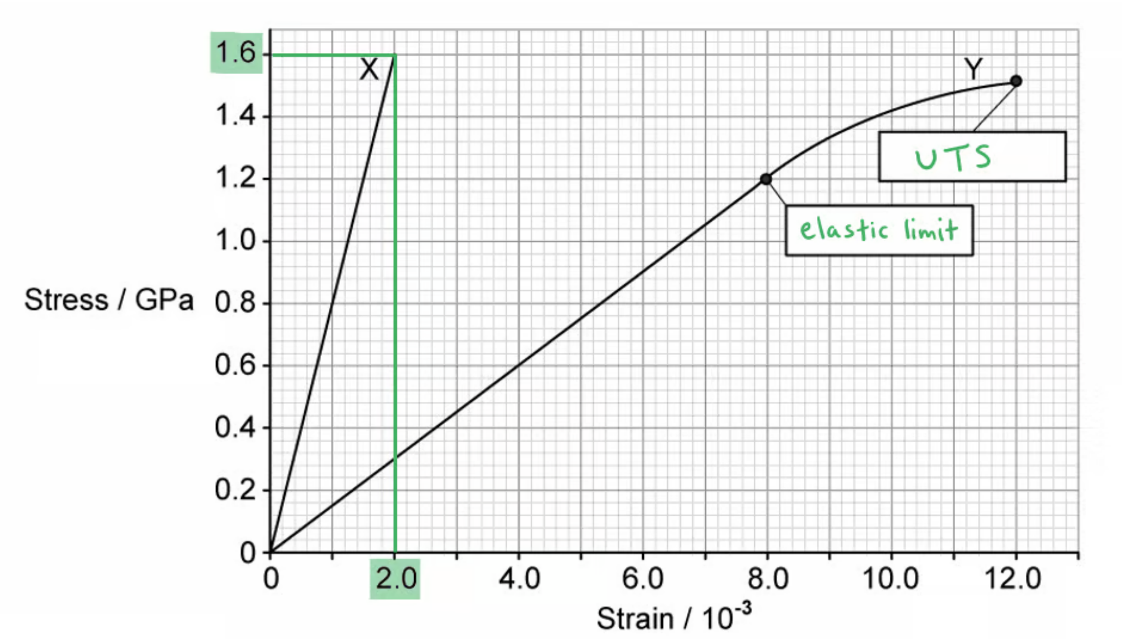
Because:

- X has the steeper gradient / larger value of stress/strain (before breaking); [1 mark]

[Total: 2 marks]

2d

(i) Determine the Young modulus of wire X:



- (From graph) when stress = 1.6 GPa, strain = 2.0×10^{-3} [1 mark]
- Young modulus = gradient of the stress-strain graph
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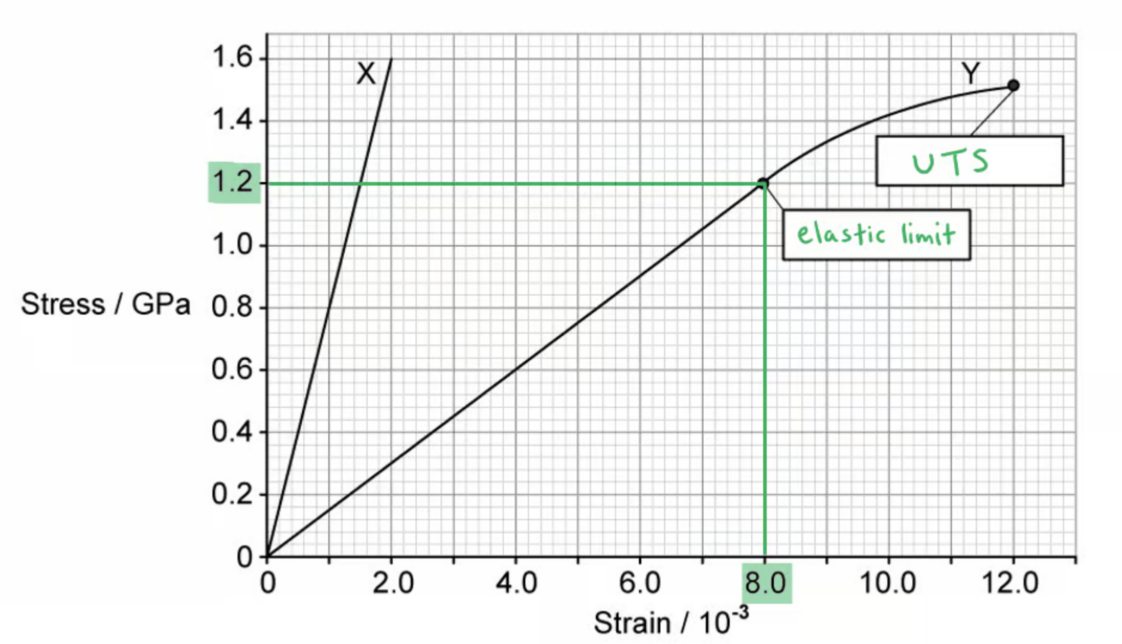
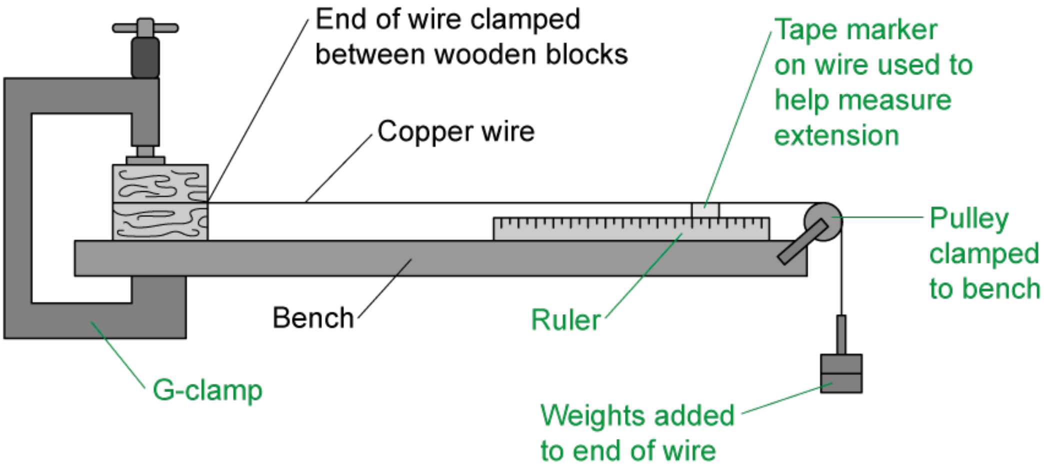


- (From graph) when stress = 1.2 GPa, strain = 8.0 × 10^{-3} [1 mark]
- Original length, L = 2.0 m
- Extension: $\varepsilon = \frac{x}{L} \quad \Rightarrow \quad x = \varepsilon L$
- Extension: $x = (8.0 \times 10^{-3}) \times 2.0$ [1 mark]
- Extension: x = 0.016 m **OR** 1.6 cm **OR** 16 mm [1 mark]

[Total: 6 marks]

3a

The correct labels are as follows:



- Tape marker / marker; [1 mark]
- Pulley; [1 mark]
- G-clamp; [1 mark]
- Ruler; [1 mark]
- Weights / masses; [1 mark]

One mark for each correct label

[Total: 5 marks]

3b

A safety consideration for this experiment is:

EITHER

- Wearing safety goggles at all times; [1 mark]
- Explanation: in case the wire snaps; [1 mark]

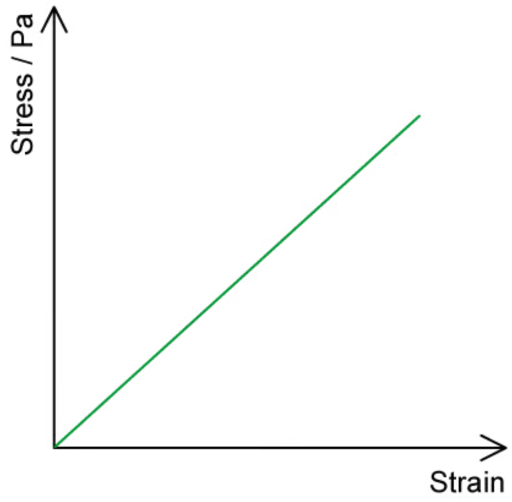
OR

- Placing a cushion / mat directly below the mass hanger; [1 mark]
- Explanation: in case the masses fall onto the ground / student's foot; [1 mark]

[Total: 2 marks]

3c

The expected variation of strain with stress for this experiment is as follows:



- Straight line drawn; [1 mark]
- Through the origin; [1 mark]

[Total: 2 marks]

The question states that the wire is not stretched beyond the limit of proportionality. This means that

- Tape marker / marker; [1 mark]
- Pulley; [1 mark]
- G-clamp; [1 mark]
- Ruler; [1 mark]
- Weights / masses; [1 mark]

One mark for each correct label

[Total: 5 marks]

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3c

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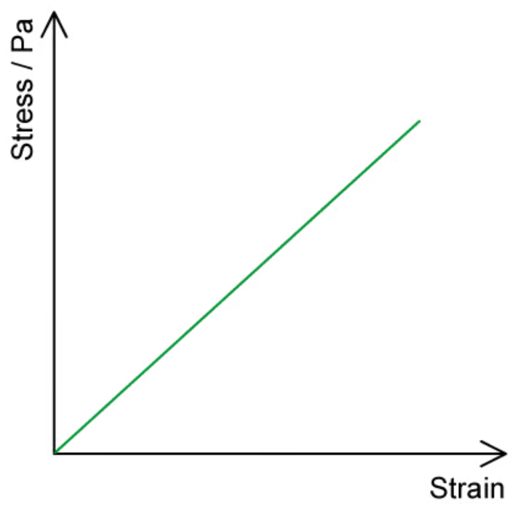

- Straight line drawn; [1 mark]
- Through the origin; [1 mark]

[Total: 2 marks]

The question states that the wire is not stretched beyond the limit of proportionality. This means that

Hooke's law is **obeyed**, therefore the stress will be **directly proportiona**l to the strain. This means the graph will be a **straight line** through the **origin**